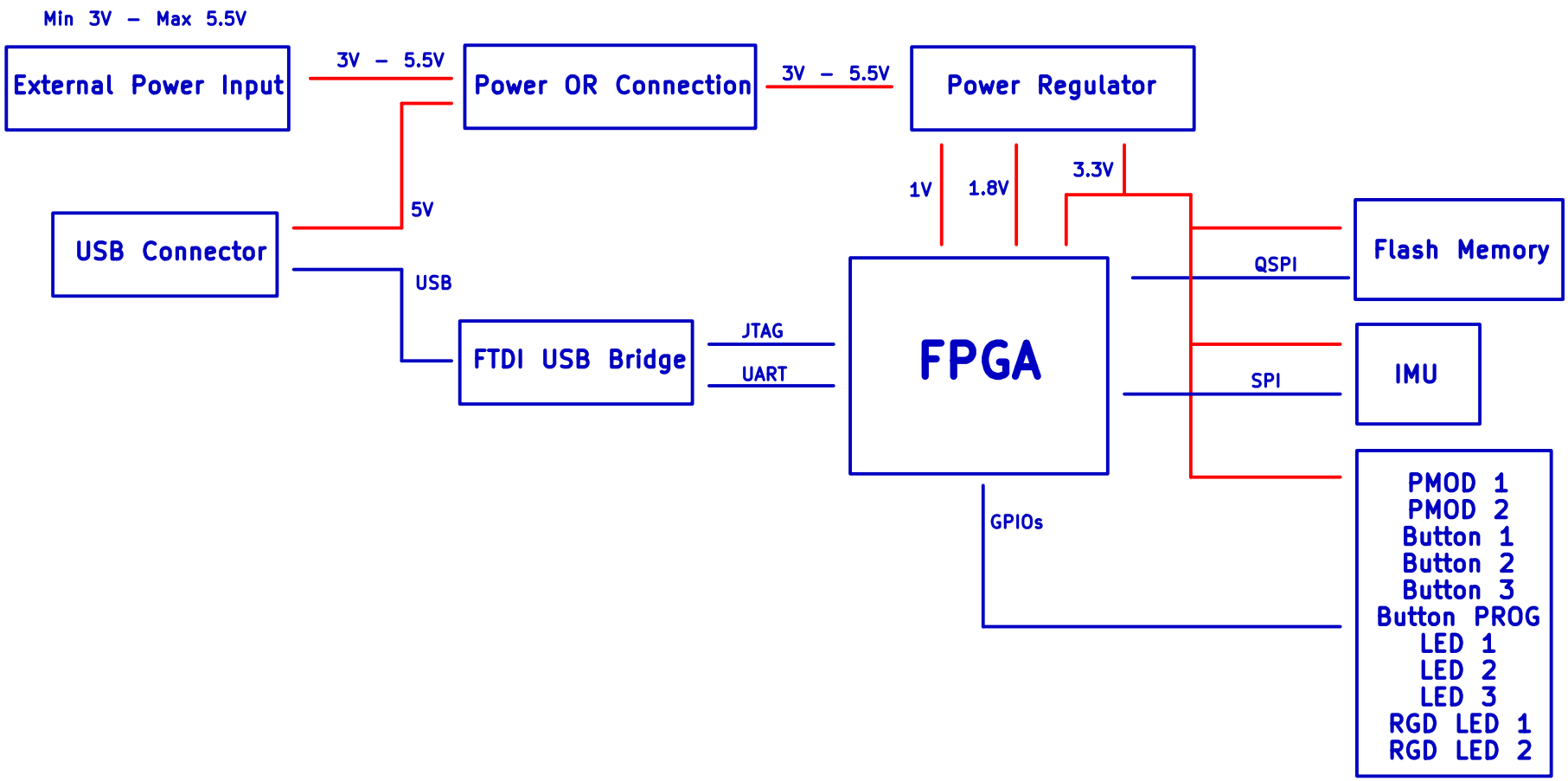
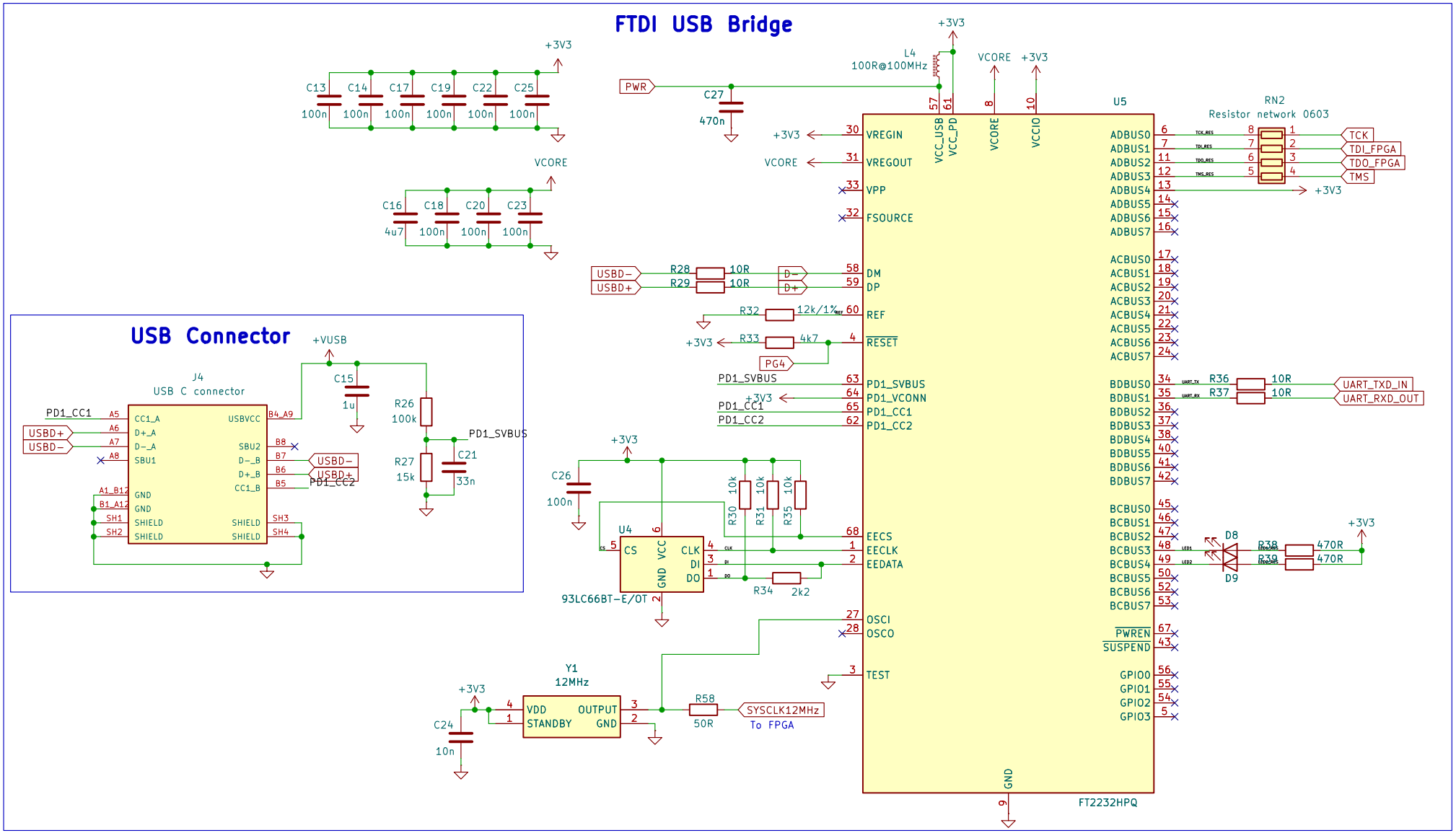
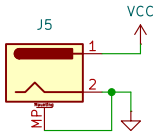
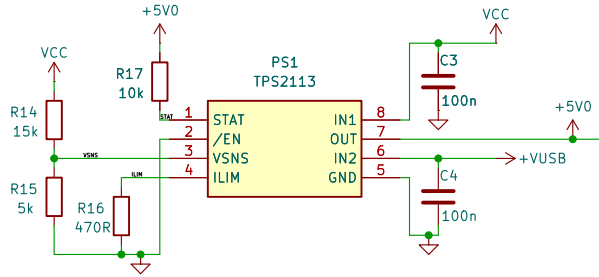

Min 3V – Max 5.5V
External Power Input
3V – 5.5V
Power OR Connection
3V – 5.5V
Power Regulator
1V
1.8V
3.3V
5V
USB Connector
USB
FTDI USB Bridge
JTAG
UART
FPGA
QSPI
Flash Memory
SPI
IMU
GPIOs
PMOD 1
PMOD 2
Button 1
Button 2
Button 3
Button PROG
LED 1
LED 2
LED 3
RGD LED 1
RGD LED 2


FTDI USB Bridge
USB Connector
J4
USB C connector
+VUSB
C15
1u
R26
100k
R27
15k
C21
33n
PD1_SVBUS
PD1_CC1
PD1_CC2
USBD+
USBD−
CC1_A
D+_A
D−_A
SBU1
USBVCC
SBU2
D−_B
D+_B
CC1_B
GND
SHIELD
+3V3
C13
C14
C17
C19
C22
C25
100n
VCORE
C16
C18
C20
C23
4u7
PWR
C27
470n
L4
100R@100MHz
VCORE
U5
FT2232HPQ
VREGIN
VREGOUT
VPP
FSOURCE
DM
DP
REF
RESET
PD1_SVBUS
PD1_VCONN
PD1_CC1
PD1_CC2
R28
R29
10R
R32
12k/1%
R33
4k7
PG4
D−
D+
U4
93LC66BT−E/OT
CS
CLK
DI
DO
VCC
GND
C26
R30
R31
R35
10k
R34
2k2
EECS
EECLK
EEDATA
OSCI
OSCO
TEST
Y1
12MHz
VDD
STANDBY
OUTPUT
R58
50R
SYSCLK12MHz
To FPGA
C24
10n
VCC_USB
VCC_PD
VCCIO
ADBUS0
ADBUS1
ADBUS2
ADBUS3
ADBUS4
ADBUS5
ADBUS6
ADBUS7
ACBUS0
ACBUS1
ACBUS2
ACBUS3
ACBUS4
ACBUS5
ACBUS6
ACBUS7
BDBUS0
BDBUS1
BDBUS2
BDBUS3
BDBUS4
BDBUS5
BDBUS6
BDBUS7
BCBUS0
BCBUS1
BCBUS2
BCBUS3
BCBUS4
BCBUS5
BCBUS6
BCBUS7
PWREN
SUSPEND
GPIO0
GPIO1
GPIO2
GPIO3
RN2
Resistor network 0603
TCK
TDI_FPGA
TDO_FPGA
TMS
R36
R37
UART_TXD_IN
UART_RXD_OUT
D8
D9
R38
R39
470R

External Power Input



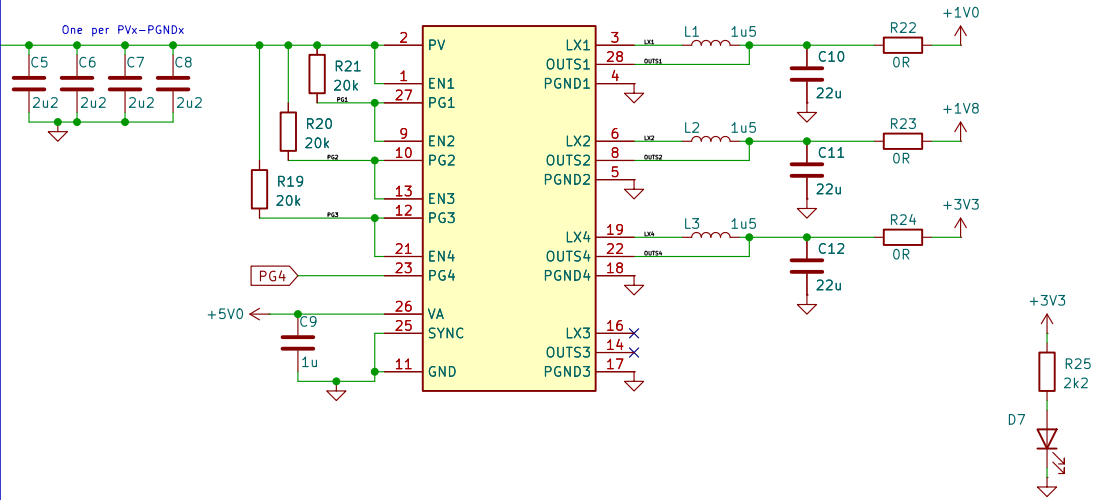
Power OR connection



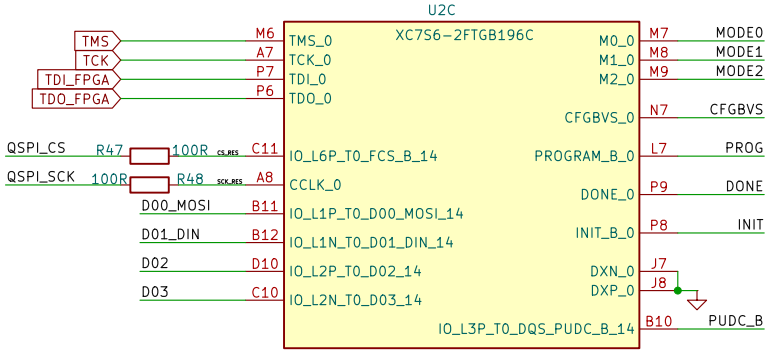
Input 2.8 – 5.5 V

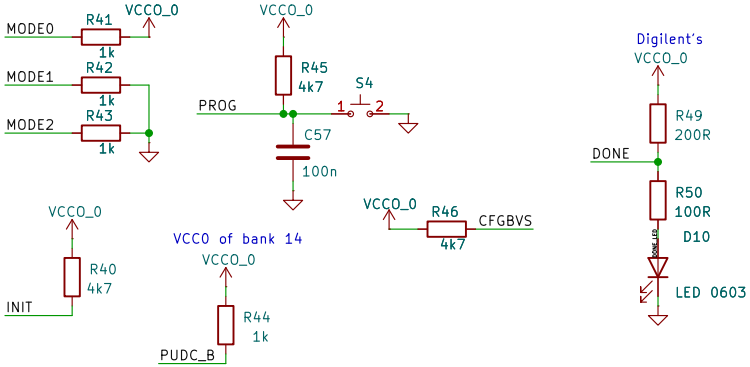
Power Regulator

Input 3 – 5.5 V

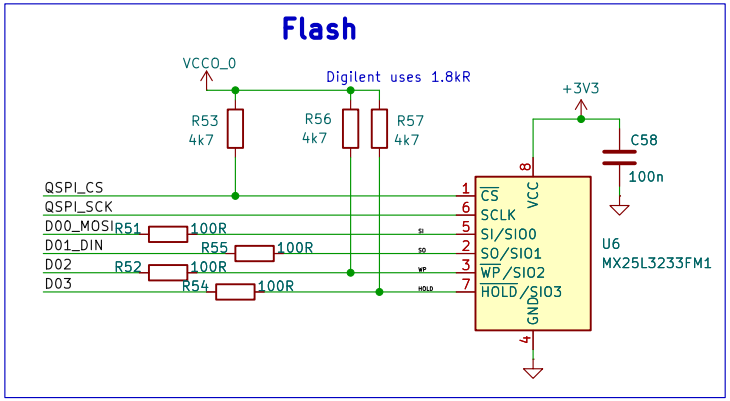


Config

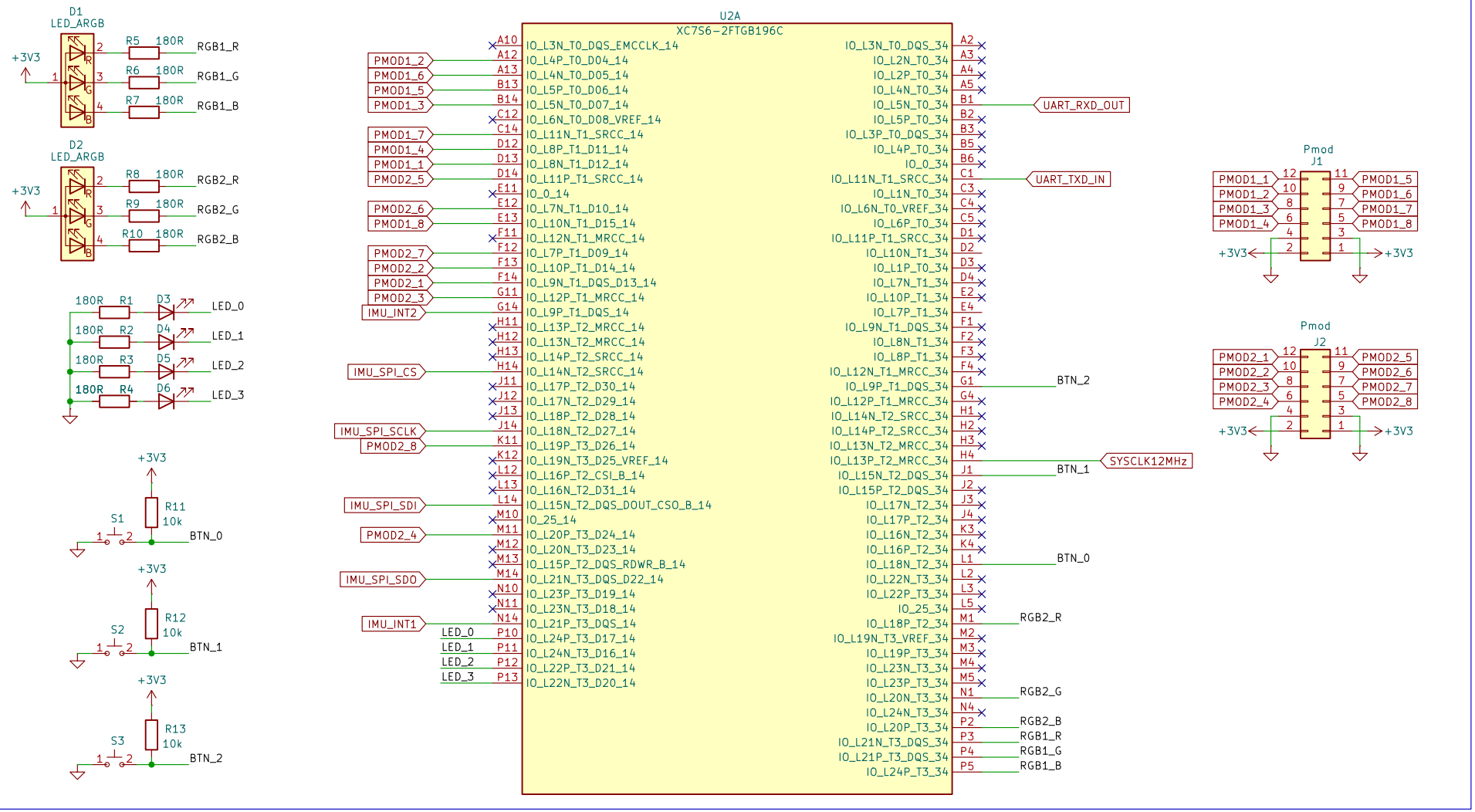




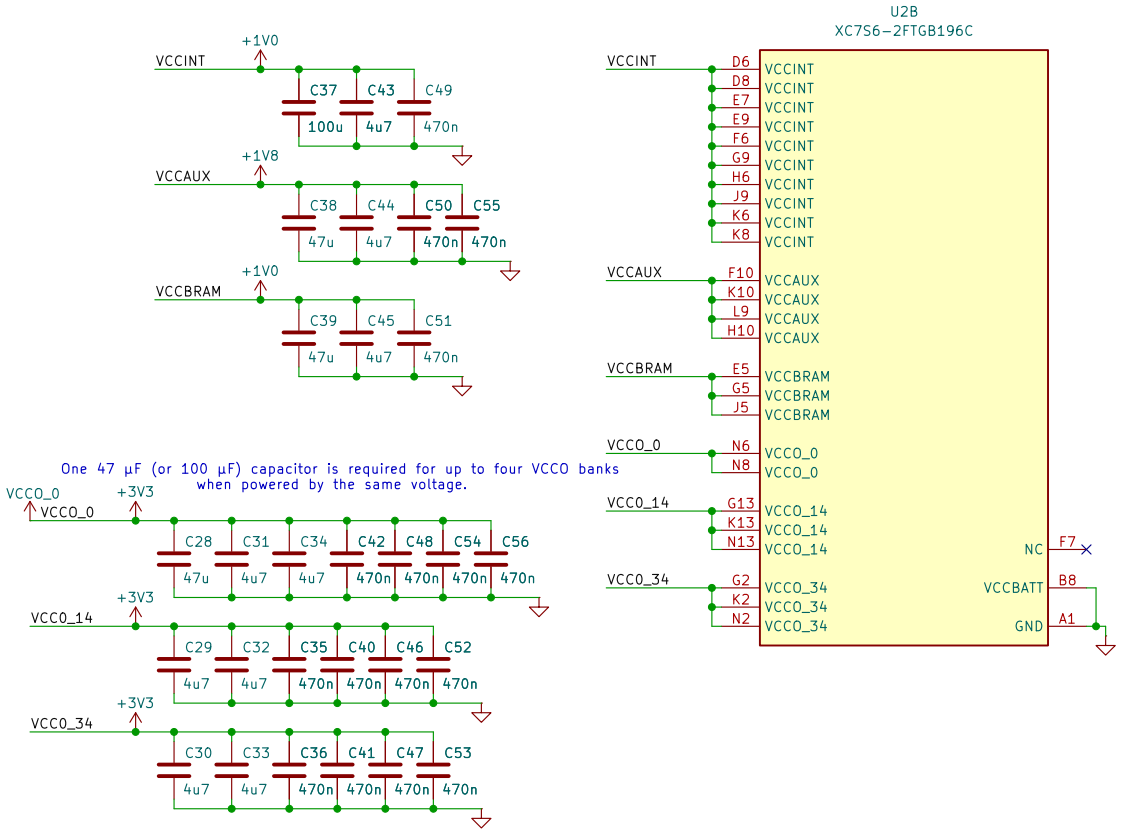
Flash

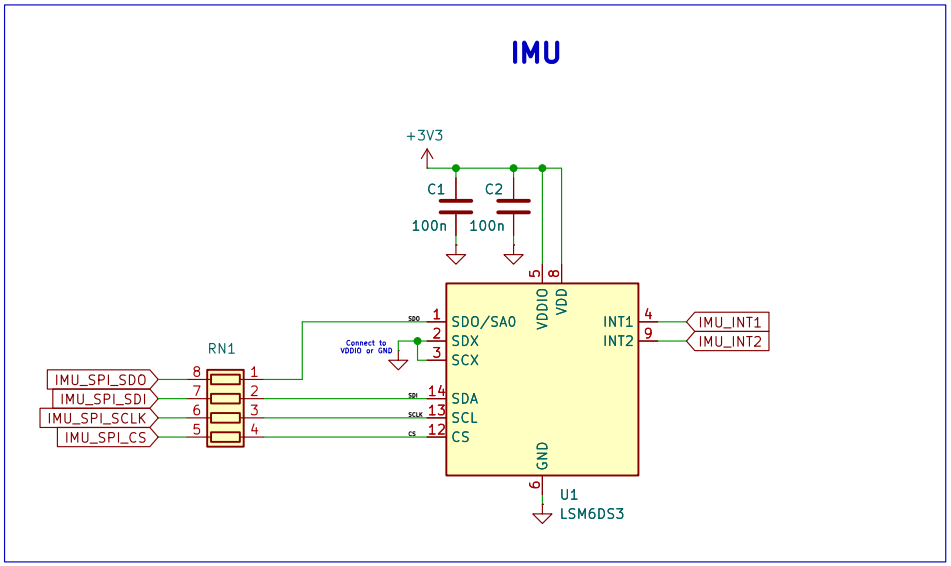


FPGA Banks Connections



FPGA Power


U2B
XC7S6-2FTGB196C
+1V0
VCCINT
C37
C43
C49
100u
4u7
470n
+1V8
VCCAUX
C38
C44
C50
C55
47u
4u7
470n
470n
+1V0
VCCBRAM
C39
C45
C51
47u
4u7
470n
One 47 µF (or 100 µF) capacitor is required for up to four VCCO banks
when powered by the same voltage.
VCCO_0
+3V3
VCCO_0
C28
C31
C34
C42
C48
C54
C56
47u
4u7
4u7
470n
470n
470n
470n
+3V3
VCCO_14
C29
C32
C35
C40
C46
C52
4u7
4u7
470n
470n
470n
470n
+3V3
VCCO_34
C30
C33
C36
C41
C47
C53
4u7
4u7
470n
470n
470n
470n
VCCINT
D6
D8
E7
E9
F6
G9
H6
J9
K6
K8
VCCINT
VCCAUX
F10
K10
L9
H10
VCCAUX
VCCBRAM
E5
G5
J5
VCCBRAM
VCCO_0
N6
N8
VCCO_0
VCCO_14
G13
K13
N13
VCCO_14
VCCO_34
G2
K2
N2
VCCO_34
NC
F7
VCCBATT
B8
GND
A1


IMU
+3V3
C1
100n
C2
100n
5
8
VDDIO
VDD
SDO/SA0
SDX
SCX
SDA
SCL
CS
GND
INT1
INT2
4
9
IMU_INT1
IMU_INT2
1
2
3
14
13
12
6
U1
LSM6DS3
Connect to VDDIO or GND
RN1
IMU_SPI_SDO
IMU_SPI_SDI
IMU_SPI_SCLK
IMU_SPI_CS
8
7
6
5
1
2
3
4